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GeoMFormer: A General Architecture for Geometric Molecular Representation Learning

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Paper Code

Problem

Molecular modeling must respect physical law: predictions stay **invariant** to rotation & translation for scalars (energy), **equivariant** for vectors (forces).

> *No framework learns both invariant and equivariant features well at once.*

Motivation

- **Hand-crafted equivariant modules.** Prior geometric nets rely on costly, specialized operations that scale poorly or sacrifice expressive power.
- **Constrained by design.** The equivariance constraint limits available operations, so architectures grow complex and hard to scale.
- **One model, both jobs.** Real applications need one network strong at both prediction types.

PRIOR GEOMETRIC NETS

Hand-built equivariant modules; scale poorly.

vs.

GEOMFORMER

Standard Transformer blocks; one scalable principle.

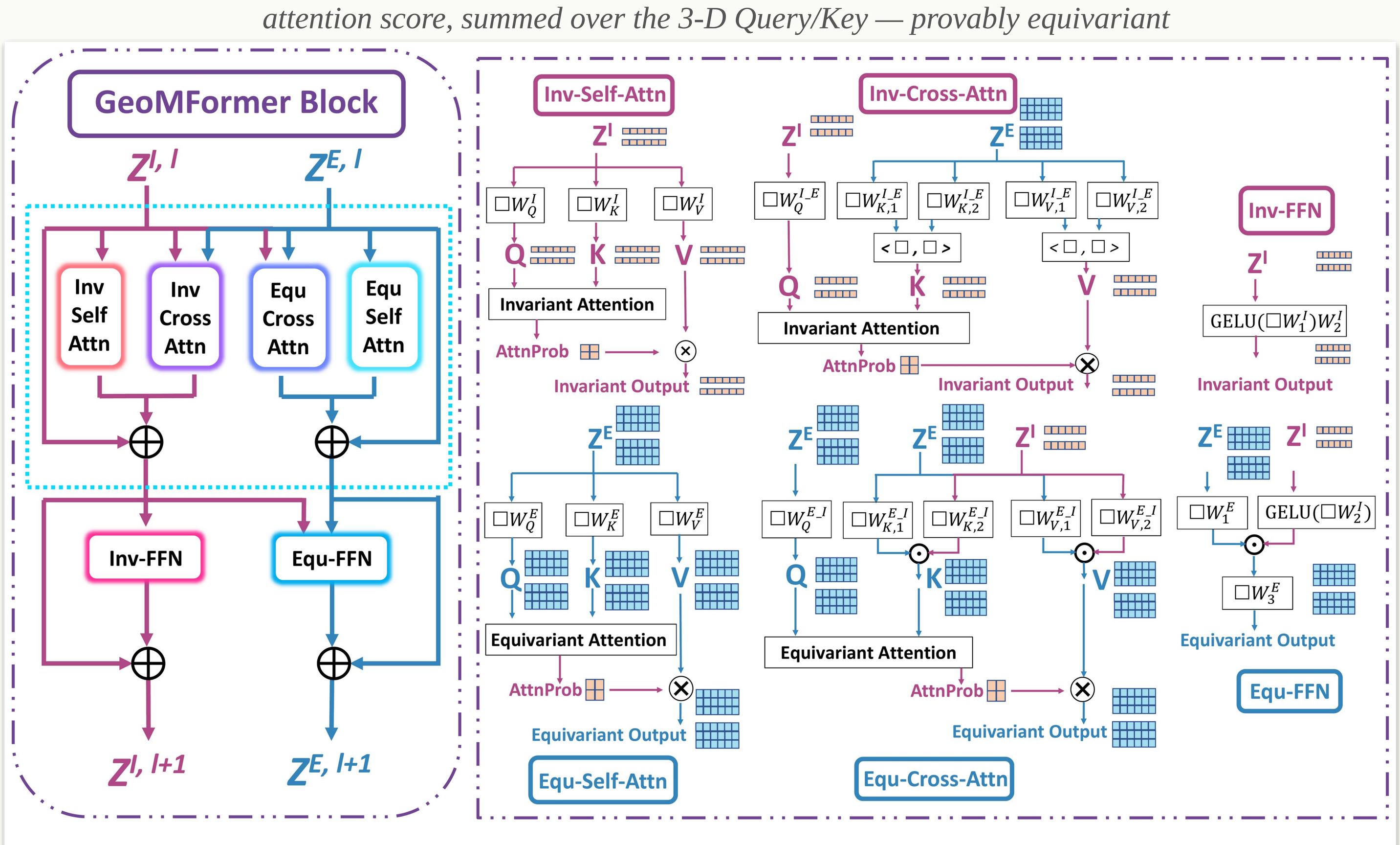
Method

- **Two parallel streams.** Every atom carries an **invariant** feature and an **equivariant** 3D feature, each in its own Transformer stream.
- **Cross-attention bridge.** Each stream queries the other (Inv←Equ, Equ←Inv), fusing the two kinds of geometric information — the key innovation.
- **Equivariance by construction.** Equivariant attention sums dot products over the 3D Query/Key, provably preserving equivariance; self-attention + FFN complete each block.
- **One general framework.** Many prior geometric models arise as special cases by restricting these modules.

$$\alpha_{ij} = \sum_{k=1}^{d} Q^{E\,\top}_{[i,:,k]} K^{E}_{[j,:,k]}$$

equivariant

attention score, summed over the 3-D Query/Key — provably equivariant



The GeoMFormer block: parallel invariant / equivariant Transformer streams bridged by self- and cross-attention.

Dataset / Benchmark

- A broad suite stressing **both** invariant (energy, HOMO-LUMO gap) and equivariant (structure, position, force) prediction.

OC20 · 460K | PCQM4Mv2 · 3.37M | Molecule3D · 2.34M | N-body sim | MD17 forces

Key Results

GeoMFormer sets state-of-the-art or highly competitive results on every benchmark, invariant and equivariant alike:

BENCHMARK (METRIC ↓)	PREV. BEST	GEOMFORMER
OC20 IS2RE · MAE (eV)	0.4410	0.4141
PCQM4Mv2 · Valid MAE	0.0867	0.0734
Molecule3D · MAE (rand.)	0.0301	0.0252
Molecule3D · MAE (scaf.)	0.1182	0.1045
N-body · MSE	**0.0071**	**0.0047**

New SOTA on OC20, Molecule3D, PCQM4Mv2 & N-body — all at O(n²) cost.

10.67 OC20 IS2RS ADwT (equivariant) vs 8.36 prior best — also wins at structure prediction

SO WHAT → *A single Transformer wins on both invariant and equivariant tasks at O(n²) complexity.*

Ablation Study

- Removing the cross-attention bridge degrades every task sharply — it is the heart of the design.

+20.8% MD17 energy | **+60.8%** MD17 forces | **+17.5%** N-body MSE

SO WHAT → *Relative gains from adding both cross-attention modules; each of the four attention blocks contributes.*

Headline Numbers

−33.8%
N-body MSE
0.0071 → 0.0047

0.0734 PCQM4Mv2 MAE

−16.3% Molecule3D MAE

+60.8% MD17 force (abl.)

Takeaway

Two standard Transformer streams — one invariant, one equivariant — bridged by cross-attention form a single general architecture that learns both representations and beats specialized geometric models.

Many prior geometric models are special cases of GeoMFormer — a unifying design principle.